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Carrier-Aware Sound Zone Control for Parametric Array Loudspeakers

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Abstract

Parametric array loudspeakers (PALs) exploit ultrasonic carrier and nonlinear self-demodulation to generate directional speech and audio sound, offering advantages for spatial audio reproduction. While recent work extends linear-acoustic sound zone control (SZC) to PAL arrays by optimizing sideband excitations and achieves high acoustic contrast (AC), the carrier—essential for demodulation—remains unexplored as a control variable. We propose a carrier-aware SZC framework for PALs, treating the carrier as a new control dimension, with an alternating optimization for sideband and carrier weights. Simulations and experiments show that our method outperforms sideband-only control, yielding higher AC and demodulation efficiency. In experiments, broadband gains of 11.2 dB in AC and 7.0 dB in target-zone SPL are achieved under identical power constraints. The results establish the carrier as a key control variable for more efficient, high-performance spatial audio and speech rendering with PALs.

Index Terms: Spatial audio, sound zone control, parametric array loudspeaker, acoustic contrast control

1. Introduction

Due to the ability to generate highly directional audio beams from a compact-size sound source, parametric array loudspeakers (PALs) have gained considerable attention in various audio applications such as spatial sound reproduction [1–3], active noise control [4, 5], and private speech communication systems [6, 7]. Compared to traditional electrodynamic loudspeaker (EDL) arrays, PAL arrays offer advantages including smaller source size and enhanced directivity for sound zone control (SZC), in which the goal is to create bright zones with desired audio content and dark zones with minimal sound leakage within a shared environment [8–16].

Conventional SZC with EDL arrays relies on the linear-acoustic framework, where the sound field can be expressed as a linear superposition of individual channel responses. However, SZC with PAL arrays is challenging due to the inherent nonlinear demodulation process. When a PAL emits an ultrasonic carrier modulated by an audio signal, the audio sound is self-demodulated due to nonlinear interactions between the carrier and sideband. Thus, interactions between carriers and sidebands from different channels produce coupled inter-channel audio components [17–20], preventing the total audio sound field from being expressed as a direct linear sum of individual channel contributions, and making conventional linear SZC framework unsuitable for direct application to PAL arrays.

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Recent research has explored several strategies to address the challenges of implementing SZC with PAL arrays. Zhuang et al. developed a framework that models the nonlinear interactions between any two channels using a pairwise transfer function matrix [21]. While effective in principle, this approach requires characterizing all channel pairs, which becomes increasingly impractical for large-scale arrays. Alternatively, Zhu et al. proposed a linearizing framework by introducing the "effective transfer function" for PAL arrays [19]. To measure the effective transfer function of a specific channel, unmodulated carrier signals in other channels are emitted simultaneously. This captures the nonlinear coupling effects caused by interactions with all channels. Thus, the "effective transfer function" of different channels can be linearly superimposed, allowing conventional linear-acoustics-based algorithms to be directly applied to PAL arrays. However, this framework focuses primarily on sideband control and treats the carrier as fixed. In PAL systems, the carrier plays a central role in governing the nonlinear demodulation process that generates the audible sound field. Fixing the carrier therefore restricts the available control degrees of freedom, which can lead to both degraded sound zone performance and reduced demodulation efficiency.

We propose a carrier-aware sound zone control framework, in which the carrier is explicitly incorporated as an active control variable alongside the sideband. An alternating optimization strategy is developed to sequentially updates the sideband and carrier excitation. As a proof of concept, this carrier-aware framework is applied to acoustic contrast control (ACC) in both simulations and experiments. Simulations are conducted using the k-space approach [22], which is a recent work that addresses the longstanding challenges in efficient and accurate simulation of multi-channel PALs caused by nonlinear demodulation and inter-channel coupling. The results show that the proposed method consistently outperforms sideband-only control, yielding higher zone contrast. Furthermore, better audio demodulation efficiency is achieved, which is especially important given PALs' inherent low-frequency reproduction inefficiency.

2. Methods

2.1. Linear Sound Zone Control Framework

The conventional SZC is achieved by optimizing the frequency-dependent weights $\mathbf{w}(\omega) = [w_1(\omega), w_2(\omega), \ldots, w_L(\omega)]^\top$ for an L-channel loudspeaker array, where $w_l(\omega)$ represents the complex weight of the l-th source channel at the angular frequency ω. For an EDL array, the total sound field can be represented in frequency domain as the superposition of sound fields generated by individual channels, $p(\mathbf{r}) = \sum_{l=1}^{L} G_l^{\mathbf{r}} w_l$,

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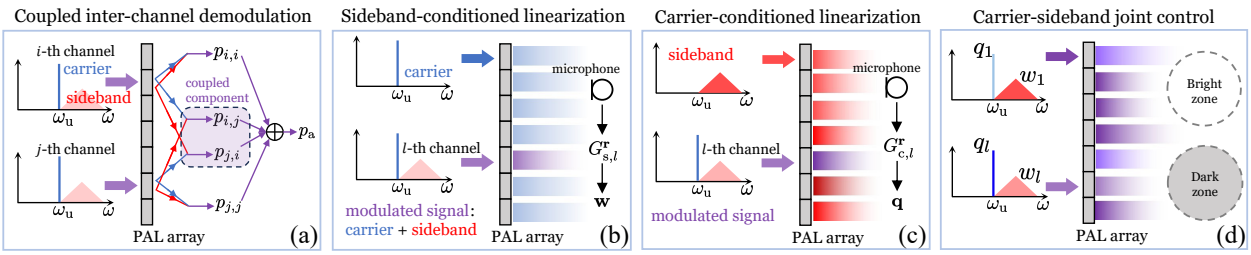


Figure 1: *Conceptual workflow of carrier-aware SZC for PAL arrays. (a) Illustration of audible sound generation via nonlinear interactions between the i-th and j-th channels; (b,c) Sideband- and carrier-conditioned linearizations, where $\mathbf{G}_{\mathrm{s}}(\mathbf{q})$ and $\mathbf{G}_{\mathrm{c}}(\mathbf{w})$ are measured by fixing the carrier or sideband as the background field, enabling alternating optimization of sideband weights $\mathbf{w}$ and carrier weights $\mathbf{q}$; (d) Final carrier-sideband joint control with $\mathbf{w}$ and $\mathbf{q}$ to form a bright zone and suppress leakage in a dark zone.*

where $G_l^{\mathbf{r}}$ is the acoustic transfer function from the l-th channel to the field point $\mathbf{r}$. With M_{b} control points in the bright zone and M_{d} points in the dark zone at positions $\mathbf{r}_{\mathrm{b}}^{m_{\mathrm{b}}}$ and $\mathbf{r}_{\mathrm{d}}^{m_{\mathrm{d}}}$ ($m_{\mathrm{b}} = 1,\ldots,M_{\mathrm{b}}, m_{\mathrm{d}} = 1,\ldots,M_{\mathrm{d}}$), the complex sound pressure vectors at these control points can then be written as $\mathbf{p}_{\mathrm{b/d}} = [p(\mathbf{r}_{\mathrm{b/d}}^1),\ldots,p(\mathbf{r}_{\mathrm{b/d}}^{M_{\mathrm{b/d}}})]^\top = \mathbf{G}_{\mathrm{b/d}}\mathbf{w}$, where $\mathbf{G}_{\mathrm{b/d}}$ is the transfer matrix for the bright or dark zone, with elements $(\mathbf{G}_{\mathrm{b/d}})_{m,l} = G_l^{\mathbf{r}_{\mathrm{b/d}}^m}, (m = 1,\ldots,M_{\mathrm{b/d}}, l = 1,\ldots,L)$.

Based on these transfer matrices, typical SZC strategies, such as ACC [8] and pressure matching [9], formulate specific objective functions to regulate the spatial sound distribution. In ACC, the goal is to maximize the acoustic contrast (AC)

$$\mathrm{AC} = \frac{\mathbf{w}^H\mathbf{R}_{\mathrm{b}}\mathbf{w}}{\mathbf{w}^H\mathbf{R}_{\mathrm{d}}\mathbf{w}}, \quad \mathbf{R}_{\mathrm{b/d}} = \mathbf{G}_{\mathrm{b/d}}^H\mathbf{G}_{\mathrm{b/d}}. \tag{1}$$

Accordingly, the classical ACC problem is formulated as [8]:

$$\max_{\mathbf{w}} \frac{\mathbf{w}^H\mathbf{R}_{\mathrm{b}}\mathbf{w}}{\mathbf{w}^H\mathbf{R}_{\mathrm{d}}\mathbf{w} + \delta\mathbf{w}^H\mathbf{w}}, \tag{2}$$

where a regularization term is typically included to improve robustness and $\delta > 0$ is the regularization factor. The optimal solution is obtained as the generalized eigenvector corresponding to the largest eigenvalue of the matrix pair $(\mathbf{R}_{\mathrm{b}}, \mathbf{R}_{\mathrm{d}} + \delta\mathbf{I})$.

2.2. Linearized Sound Zone Control for PAL Arrays with Carrier Awareness

With the commonly used upper-sideband amplitude modulation, the modulated signal emitted by the PAL is written as

$$y(t) = \cos(\omega_{\mathrm{u}}t) + \big[s(t)\cos(\omega_{\mathrm{u}}t) - \hat{s}(t)\sin(\omega_{\mathrm{u}}t)\big], \tag{3}$$

where ω_{u} is the ultrasonic carrier's angular frequency, $s(t)$ is the audio signal to be reproduced, and $\hat{s}(t)$ is its Hilbert transform. The two terms on the right-hand side correspond to the carrier and sideband components, respectively. The audio sound is self-demodulated via the nonlinear interaction between carriers and sidebands. Since interactions exist between all channels, as illustrated in Fig. 1(a), the total sound pressure is the sum of each pair of channels [18–20]:

$$p_{\mathrm{a}} = \sum_{i,j=1}^{N} p_{i,j} = \sum_{i=1}^{N} p_{i,i} + \sum_{i,j=1,i\neq j}^{N} p_{i,j}, \tag{4}$$

where the first term on the right-hand side represents the nonlinear self-interaction within each channel (between its own carrier and sidebands), and the second term accounts for the cross-channel interactions between different channels, which prevent the demodulated field from being a direct linear summation of individual channel contributions.

To address this, we first employ the linearization method based on a customized sideband–conditional transfer function $\mathbf{G}_{\mathrm{s}}$ [19]. As depicted in Fig. 1(b), the l-th channel transfer function $G_{\mathrm{s},l}$ is obtained by driving only the l-th channel with the full modulated signal while all other channels simultaneously transmit pure carriers. This procedure captures the nonlinear interactions between the l-th sideband and all channels' carriers. Thus, the total audio sound could be represented as the superposition of individual channel responses, enabling the direct application of the linear SZC framework for optimizing $\mathbf{w}$.

It is noted that all the measured transfer functions correspond to the audio sound. Therefore, the optimized weights $\mathbf{w}$ act directly on the audio signal $s(t)$. However, as is seen from Eq. (3), manipulating $s(t)$ influences only the sidebands, leaving the carriers as a fixed, unoptimized background. To enable carrier optimization, we exploit the inherent symmetry of the modulation structure.

In the frequency domain, with $s(t) = \cos(\omega_{\mathrm{a}}t)$ and $\hat{s}(t) = \sin(\omega_{\mathrm{a}}t)$, Eq. (3) can be reformulated as:

$$\begin{aligned} y(t) &= \cos(\omega_{\mathrm{u}}t) + \cos[(\omega_{\mathrm{u}} + \omega_{\mathrm{a}})t] \\ &= \cos(\omega_{\mathrm{u}}'t) + \big[s(t)\cos(\omega_{\mathrm{u}}'t) + \hat{s}(t)\sin(\omega_{\mathrm{u}}'t)\big], \end{aligned} \tag{5}$$

where $\omega_{\mathrm{u}}' = \omega_{\mathrm{u}} + \omega_{\mathrm{a}}$ denotes the sideband frequency. This reformulation reveals the symmetric roles of the carrier and sideband components. Consequently, by interchanging their roles, the carrier optimization problem can be mapped onto a framework formally equivalent to sideband control.

We define a carrier-conditioned transfer function $\mathbf{G}_{\mathrm{c}}$, measured under a complementary configuration: the channel to be measured emits the full modulated signal, while the remaining channels emit sideband signals as shown in Fig. 1(c). Applying a SZC algorithm using $\mathbf{G}_{\mathrm{c}}$ yields an intermediate weight vector $\mathbf{w}^{\mathrm{c}}$ that corresponds to the audio signal. To derive the actual carrier weight $\mathbf{q}$, we map $\mathbf{w}^{\mathrm{c}}$ back to the carrier domain by analyzing the changes in amplitude and phase before and after modulation. For $w_l^{\mathrm{c}} = A\exp(\mathrm{i}\phi)$, the audio signal is $s(t) = A\cos(\omega_{\mathrm{a}}t + \phi)$ and its Hilbert transform is $\hat{s}(t) = A\sin(\omega_{\mathrm{a}}t + \phi)$. Substituting them into Eq. (5) gives:

$$\begin{aligned} y(t) &= \cos[(\omega_{\mathrm{u}} + \omega_{\mathrm{a}})t] + \big[A\cos(\omega_{\mathrm{a}}t + \phi)\cos(\omega_{\mathrm{u}}t) \\ &\quad + A\sin(\omega_{\mathrm{a}}t + \phi)\sin(\omega_{\mathrm{u}}t)\big] \\ &= \cos[(\omega_{\mathrm{u}} + \omega_{\mathrm{a}})t] + A\cos(\omega_{\mathrm{u}}t - \phi), \end{aligned} \tag{6}$$

which shows that the complex weight of the carrier is $q_l = A\exp(-\mathrm{i}\phi)$, the complex conjugate of w_l^{c}, yielding $\mathbf{q} = \mathbf{w}^{\mathrm{c},*}$, where $*$ denotes conjugation.

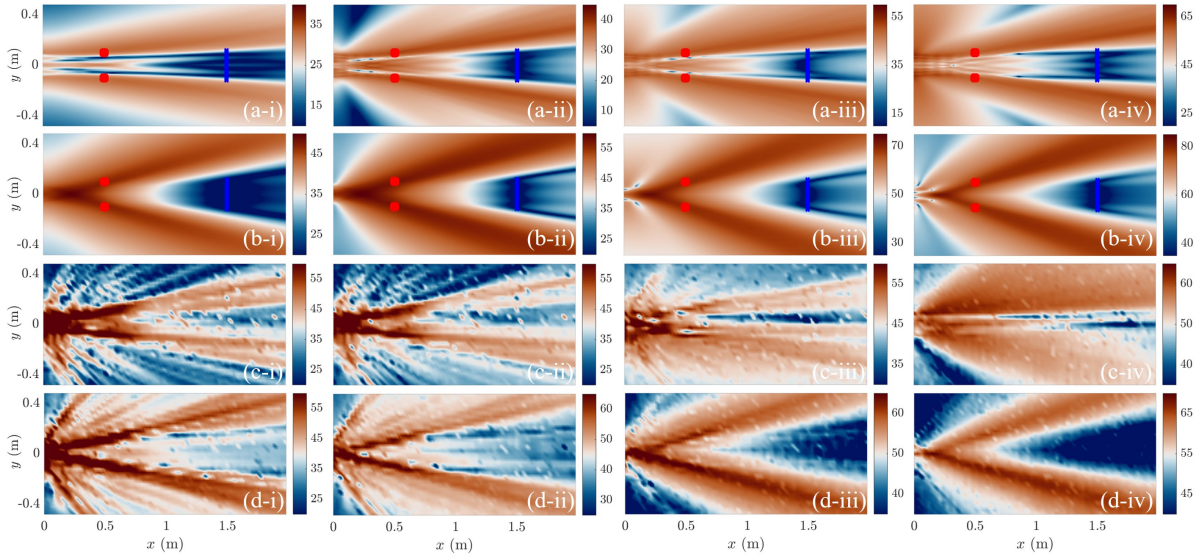


Figure 2: *Simulated (a,b) and measured (c,d) audio SPL distributions. (a,c): sideband-only ACC, (b,d): carrier-aware ACC. (i) to (iv): 500 Hz, 1 kHz, 2 kHz, 4 kHz. The red (blue) markers at $x = 0.5$ m ($x = 1.5$ m) are bright (dark) zone control points.*

Consequently, the total audio sound field can be represented through two distinct linearized operators:

- **Sideband operator** $\mathbf{G}_{\mathrm{s}}(\mathbf{q})$: given fixed carrier weights $\mathbf{q}$, the demodulated field behaves linearly with respect to $\mathbf{w}$.
- **Carrier operator** $\mathbf{G}_{\mathrm{c}}(\mathbf{w})$: given fixed sideband weights $\mathbf{w}$, the demodulated field exhibits a linear dependence on carrier weights $\mathbf{q}$.

2.3. Alternating Carrier-Aware Optimization Strategy

As the effective transfer functions are role-dependent, the proposed carrier-aware control here is realized through a finite-stage alternating re-linearization and optimization procedure, in which the sideband and carrier are optimized sequentially:

1. Initial Sideband Optimization: Given an initial carrier configuration $\mathbf{q} = [1, 1, \ldots, 1]^{\top}$, the sideband-conditioned transfer function $\mathbf{G}_{\mathrm{s}}(\mathbf{q})$ is measured. The sideband weights $\mathbf{w}$ are then optimized using a standard SZC algorithm.
2. Carrier Optimization: With the optimized sideband weights $\mathbf{w}$ fixed, the system is re-linearized by measuring the carrier-conditioned transfer function $\mathbf{G}_{\mathrm{c}}(\mathbf{w})$. The carrier weights $\mathbf{q}$ are then updated by the same SZC formulation.
3. Sideband Re-alignment: Since the ultrasonic carrier is a single-frequency component, the carrier optimization in Stage 2 is performed with respect to a specific audio frequency. As a result, the optimized carrier weights are inherently frequency-specific and may not be optimal across a wide audio bandwidth. Therefore, to ensure the sidebands are perfectly matched to this refined carrier background, the audio weights $\mathbf{w}$ are re-optimized in this final stage using the updated sideband-conditioned transfer matrix $\mathbf{G}_{\mathrm{s}}(\mathbf{q})$.

In this way, we obtain both $\mathbf{w}$ and $\mathbf{q}$ to control the sideband and the carrier respectively, resulting the carrier-aware SZC for a PAL array as depicted in Fig. 1(d). Since the transfer functions are empirically measured, environmental reflections are naturally encapsulated, making the framework theoretically applicable to arbitrary acoustic environments including indoor spaces.

In principle, this alternating process could be repeated. However, in practice, each stage requires a complete re-measurement of the transfer functions (either $\mathbf{G}_{\mathrm{s}}$ or $\mathbf{G}_{\mathrm{c}}$), which imposes a significant experimental and temporal cost. Therefore, rather than iterating to convergence, this work adopts a pragmatic, validation-oriented perspective by employing this finite-stage procedure that captures the primary benefits of carrier optimization while maintaining high operational efficiency.

3. Simulations and experiment validations

Simulations and experiments are conducted to validate the proposed carrier-aware SZC framework. As a proof-of-concept implementation, ACC is adopted as the sound zone optimization method, allowing a direct comparison between conventional sideband-only control and the proposed carrier-aware approach.

The control point configuration utilized in both simulations and experiments is illustrated in Fig. 2, where bright and dark zone control points are marked in red and blue, respectively. The bright zone comprises two clusters of 9 points each, centered at $(0.5\,\mathrm{m}, \pm 0.1\,\mathrm{m})$ in a 3×3 grid with 1 cm spacing. The dark zone includes 25 points uniformly distributed along a line with 1 cm spacing, centered at $(1.5\,\mathrm{m}, 0)$ and spanning a total length of 24 cm. This configuration is designed to emulate a typical personal audio scenario, where the bright zone corresponds to the listener's ear region in front of the PAL, and the dark zone is the rear region intended for acoustic suppression.

Both simulations and experiments employ the same PAL array configuration. As shown in Fig. 3, the array consists of 24 columns, each with 24 circular emitters of 5 mm radius. The columns are arranged compactly with a center-to-center spacing of 8.7 mm. Due to the symmetric control point distribution, a symmetric channel design is adopted, pairing two columns symmetric about the central axis ($y = 0$) as one channel. The ultrasonic carrier frequency is 40 kHz, and the carrier optimization stage is performed at an audio frequency of 1 kHz.

Experimental measurements are conducted in an anechoic chamber as shown in Fig. 3. A condenser microphone Type CY-407 (Hangzhou Acoustic Sensing Technology Co., Ltd., Hangzhou, China) is used to measure the sound pressure, which is covered with a thin piece of plastic film when measuring audio sound to mitigate spurious sound [23, 24]. The sound field over a 2 m × 1 m plane is scanned using a stepper-motor-driven measurement system with a spatial resolution of 3 cm.

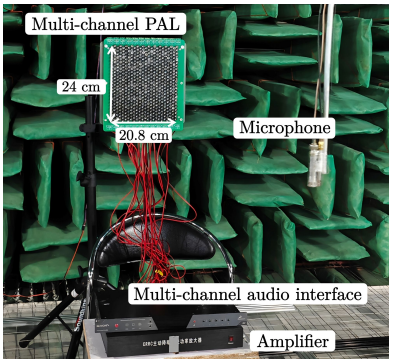


Figure 3: *A photo of the experimental setup.*

Figure 2 presents the simulated and measured audio sound pressure level (SPL) distributions for sideband-only ACC and carrier-aware ACC at 500 Hz, 1 kHz, 2 kHz, and 4 kHz. For sideband-only ACC, shown in Fig. 2(a,c), although the sound energy is successfully steered toward the bright-zone control points, a substantial amount of sound leakage toward the dark zone remains observable. This leakage becomes more pronounced at higher frequencies, indicating that suppressing the demodulated audio field in the dark zone is fundamentally constrained when only the sideband is controlled. In contrast, the carrier-aware ACC results in Fig. 2(b,d) exhibit a clear reduction of sound leakage into the dark zone, accompanied by a enhanced sound energy concentration in the bright zone across all tested frequencies. The consistent performance improvement is observed in both simulation and experimental measurements, confirming that the advantage of the carrier-aware approach is robust against practical system imperfections.

To clarify the physical mechanism underlying the performance improvement, Fig. 4 depicts the corresponding ultrasonic SPL distributions of the carrier (40 kHz) and sideband (41 kHz) at 1 kHz audio frequency. As shown in Fig. 4(ii), by optimizing the audio-driven sideband signal, sideband-only ACC steers the ultrasonic sideband energy toward the bright zones and reduces its level in the dark zone, thus improving AC of the audio field. However, the carrier remains uncontrolled, resulting in a nearly collimated beam pattern as shown in Fig. 4(i), which limits the PAL array's ability to suppress leakage in the dark zone. Physically, the demodulated audio sound can be interpreted as radiated from a virtual source with the source density proportional to the spatial product of the carrier and sideband sound pressures. Thus, the spatial mismatch between the carrier and sideband in sideband-only ACC not only degrades the achievable dark-zone suppression but also reduces the audio demodulation efficiency, leading to a lower SPL in the bright zone. In contrast, carrier-aware ACC addresses these limitations by controlling the carrier. As depicted in Fig. 4(iii,iv), the carrier optimization in carrier-aware ACC aligns its spatial distribution with the sideband field, thereby enhancing AC and improving the SPL in the bright zone.

Broadband experiments from 500 Hz to 4 kHz are conducted using white noise as the audio signal, with time-domain control filters derived by windowing the inverse discrete Fourier transform of the optimal weights. For a fair comparison, the power constraint $\mathbf{w}^{\mathrm{H}}\mathbf{w}$ is kept identical for both control strategies. Figure 5 compares the measured broadband audio responses in the bright and dark zones for sideband-only and carrier-aware ACC, showing that carrier-aware ACC achieves consistently higher AC and bright-zone response across the entire frequency band. The total AC achieved by sideband-only

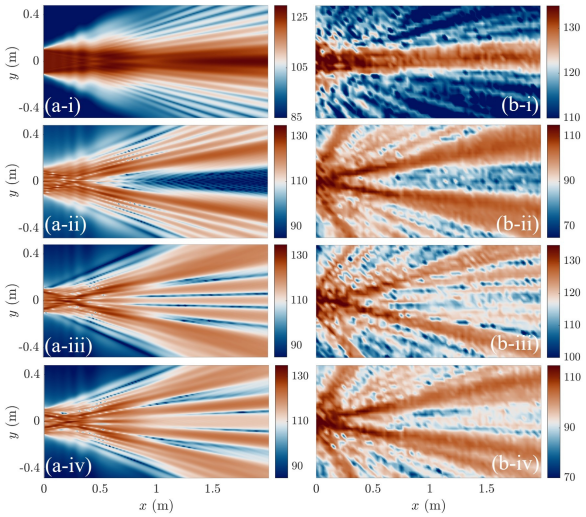


Figure 4: *Simulated and measured ultrasound SPL distributions. (a) Simulation and (b) measurement; (i,ii) sideband-only ACC, (iii,iv) carrier-aware ACC; (i,iii) carrier (40 kHz), (ii,iv) sideband (41 kHz) corresponding to 1 kHz audio sound.*

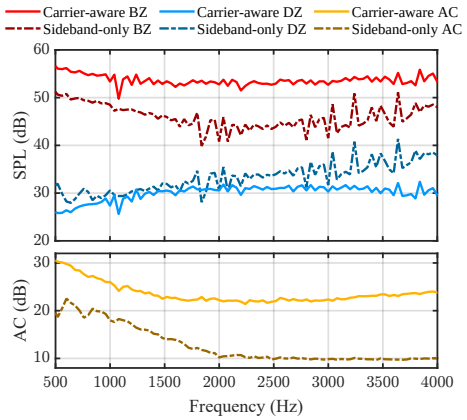


Figure 5: *Broadband performance of two strategies. BZ and DZ in the legend denote the responses of the Bright Zone and the Dark Zone, respectively.*

and carrier-aware ACC is 12.4 dB and 23.6 dB, respectively, corresponding to an improvement of 11.2 dB. Moreover, the carrier-aware ACC yields a 7.0 dB increase in total bright-zone SPL, indicating significantly higher demodulation efficiency. These results demonstrate that carrier-aware ACC consistently achieves higher AC while maintaining a higher bright-zone SPL across the entire frequency band, thereby confirming the effectiveness of the proposed carrier-aware SZC framework.

4. Conclusion

In this paper, we propose a carrier-aware SZC framework for parametric array loudspeakers. By explicitly leveraging the carrier as a control dimension rather than a fixed background component, the proposed approach overcomes a fundamental limitation of existing methods, significantly enhancing the overall performance of PAL-based SZC. Simulations and experiments based on ACC demonstrate that our method leads to a substantial increase in AC and demodulation efficiency, indicating that the carrier-aware optimization is a key enabling concept for high-performance sound zone control with PAL arrays.

5. Generative AI Use Disclosure

The authors confirm that no generative AI tools were used to produce any substantial content of this paper. Generative AI was only employed for minor editing and language polishing to improve clarity and readability. All authors remain fully responsible and accountable for the work, results, and content presented herein.

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